Electoral Equilibrium: A Stochastic Optimization Framework for Coalition Rebalancing Under Political Shocks

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Abstract

We present Electoral Equilibrium, a stochastic optimization framework that estimates how a political party's voter coalition must rebalance after an exogenous shock. Given a free-text description of a political event, the framework outputs (i) updated coalition weights over five racial/ethnic blocs, and (ii) a 90% credible interval on the post-shock probability of winning the Electoral College. The framework combines three independently validated components: a fine-tuned Mistral 7B language model that translates shock narratives into per-stratum loyalty-shift bins, a disciplined quasi-convex program (DQCP) that maximizes the probability of meeting an Electoral College threshold under a Logistic-Normal ILR uncertainty model, and an NLP pipeline (RoBERTa + SetFit) that constructs training data from social media and news archives. The demographic architecture models three independent marginal strata—race, religion, and gender—without nested cross-tabulations, acknowledging the ecological fallacy as an explicit approximation. Iterative proportional fitting (IPF) calibrates the stratum layer weights from 20 historical election cycles (1948–2024); the raked solution assigns substantially higher explanatory weight to gender ($\lambda_3 = 0.662$) than to race ($\lambda_1 = 0.114$), recovering from first principles the documented growth of the gender gap since 1980. A Gaussian Process leave-one-cycle-out classifier trained on panel vote shares achieves 85% accuracy and Brier score 0.071 on the 20-cycle panel. Electoral College asymmetry analysis places the Democrat win threshold at 50.66% of the two-party popular vote versus 49.34% for Republicans.

Keywords: coalition optimization, stochastic programming, Gaussian processes, electoral forecasting, demographic modeling, quasi-convex programming, Logistic-Normal distribution

1 Introduction

Political parties do not operate in equilibrium. Exogenous shocks—Supreme Court rulings, geopolitical crises, candidate scandals, policy reversals—shift the loyalties of demographic subgroups, sometimes dramatically and rapidly. A party's strategic challenge after such a shock is to determine how to rebalance its coalition: which groups to mobilize, which to cede, and at what allocation of organizing effort. This rebalancing problem is structurally similar to portfolio optimization under uncertainty, where the "assets" are demographic blocs, the "returns" are vote shares, and the "win condition" is an Electoral College threshold rather than a portfolio return target.

The analogy is not new. Erikson, MacKuen, and Stimson's 2002 *The Macro Polity* established that presidential approval and economic sentiment drive aggregate electoral outcomes, and a large subsequent literature has modeled vote shares as functions of economic fundamentals (Abramowitz, 2008; Hibbs, 2012). More recently, the growth of individual-level survey data has enabled multilevel regression and poststratification (MRP) approaches that decompose vote share across demographic subgroups (Gelman & Little, 1997; Park et al., 2004; Lax & Phillips, 2009). High-dimensional Bayesian hierarchical models now achieve sub-percentage-point state-level forecast accuracy in U.S. presidential elections (Linzer, 2013; The Economist, 2020).

Two gaps in this literature motivate the present work. First, existing forecasting frameworks are *predictive*: they estimate outcomes given current conditions. They do not answer the prescriptive question of how a party should rebalance its coalition to improve electoral prospects after a shock shifts the baseline. Second, the treatment of uncertainty in coalition-level models is typically frequentist— Bayesian models exist but standard Dirichlet or normal approximations to the coalition weight simplex distort the geometry of the uncertainty in ways that matter for tail-event scenarios (wave elections, demographic realignments).

We address both gaps. The Electoral Equilibrium framework answers the prescriptive question directly: given a shock and its estimated per-bloc loyalty shifts, which reallocation of organizing effort maximizes the probability of meeting the Electoral College win threshold? The answer is computed via a disciplined quasi-convex program (Diamond & Boyd, 2019) that generalizes the classical Sharpe-ratio portfolio objective to the simplex-constrained coalition setting. Uncertainty is propagated via a Logistic-Normal distribution in isometric log-ratio (ILR) coordinates (Egozcue et al., 2003), which correctly handles wave elections where all blocs move together—a case that Dirichlet sampling cannot represent because its off-diagonal covariances are by construction negative.

The framework operates as a three-stage pipeline:

1. **Shock estimation (Stage 1):** A QLoRA-fine-tuned Mistral 7B language model (Jiang et al., 2023; Hu et al., 2022) translates a free-text shock narrative into per-bloc, per-stratum loyalty-shift bins using constrained decoding (Willard & Louf, 2023).

2. **Coalition optimization (Stage 2):** A CVXPY quasi-convex program maximizes $\Phi\Big(\big(\mu_{\text{eff}}(\mathbf{w}) - V_{\text{eq}}\big)/\sqrt{\lambda_1^2\,\mathbf{w}^\top \Sigma_\Delta \mathbf{w}}\Big)$ over the probability simplex, where Φ is the standard normal CDF, μ_{eff} is the three-stratum effective loyalty scalar, V_{eq} is the EC-adjusted win threshold, and Σ_Δ is the empirical covariance of race-bloc loyalty shifts.

3. **NLP pipeline (Stage 3):** A RoBERTa sentence encoder (Liu et al., 2019) combined with a SetFit few-shot demographic classifier (Tunstall et al., 2022) produces bloc-weighted sentiment scores from social media and news archives, supplying fine-tuning training data for Stage 1.

Our primary contributions are:

1. A three-stratum additive independence architecture (race, religion, gender) calibrated by iterative proportional fitting, with explicit ecological fallacy acknowledgment and an empirical comparison of additive versus raked outputs.

2. An Electoral College–adjusted win threshold derived from logistic regression on 20 historical cycles, placing the Democrat threshold at $V_{\text{eq}} = 0.5066$ versus $V_{\text{eq}} = 0.4934$

for Republicans.

3. A quasi-convexity proof for the Sharpe-ratio coalition objective and a demonstration that min-variance optimization—the standard portfolio baseline—systematically assigns zero weight to Latino and Asian blocs while the DQCP optimizer restores positive weights in deficit scenarios.

4. Empirical validation of the GP baseline classifier (85% LOCO accuracy, Brier 0.071) and documentation of three data-quality issues—Perot three-party contamination, ANES oversampling in 1988, and pre-realignment era drift—whose corrections non-trivially affect downstream estimates.

The remainder of the paper proceeds as follows. Section 2 reviews related work. Section 3 describes the framework architecture. Section 4 details the panel construction. Section 5 presents the methodology for each component. Section 6 reports empirical findings. Section 7 discusses limitations, the ecological fallacy, and calibration against prediction markets. Section 8 concludes.

2 Background and Related Work

2.1 Electoral forecasting

Quantitative electoral forecasting in the United States divides broadly into structural models, which predict outcomes from macroeconomic and political "fundamentals" (Fair, 1978; Abramowitz, 2008; Hibbs, 2012), and polling-aggregation models, which update structural priors with survey data (Erikson & Wlezien, 2014; Linzer, 2013). A third class uses individual-level panel surveys via MRP to produce small-area estimates (Gelman & Little, 1997; Lax & Phillips, 2009; Wang et al., 2015). All three classes are primarily predictive; none frames the coalition rebalancing question as an optimization problem.

The closest antecedents to our work are the "margin of error" decomposition models (King, 1991) and simulation-based approaches to electoral uncertainty (Jackman, 2005; Strauss, 2007). Gaussian Process models for vote shares appear in Linzer (2013) and Lock & Gelman (2010), who use GP priors over election-day outcomes. Our use of GPR as a baseline win-probability classifier over 20-cycle panel data differs in purpose: we are not forecasting an upcoming election but validating that our panel vote-share features are informative about realized outcomes, which is the necessary precondition for the optimizer's objective.

2.2 Portfolio theory and coalition modeling

The analogy between political coalitions and investment portfolios has been noted informally by practitioners but formally developed by only a handful of political economists. Enelow & Hinich (1984) established spatial models of coalition building. Closer to our framing, Erikson & Wlezien (2012) used a variance-minimization framing to model party positioning. The min-variance interpretation fails in exactly the deficit scenarios we target: a party whose effective loyalty is below the win threshold cannot improve its probability of winning by minimizing variance—it must maximize the probability of the tail event (Roy, 1952).

The Sharpe-ratio objective we employ has antecedents in financial portfolio theory (Sharpe, 1966) and in reliability engineering (Roy, 1952). Its adaptation to the quasi-convex programming setting via DQCP is due to Diamond & Boyd (2019), who prove that a ratio of an affine function to a positive convex function is quasi-convex and can be solved exactly by disciplined convex programming software such as CVXPY.

2.3 Compositional data and the simplex

Coalition weight vectors lie on the probability simplex. Standard normal or Dirichlet approximations distort the geometry of uncertainty in compositional data in well-documented ways (Aitchison, 1986). The Dirichlet distribution forces all off-diagonal covariances negative, making it impossible to model the positive correlations that characterize wave elections. The delta method with a floor constraint distorts the Aitchison geometry non-linearly near the boundary (Egozcue et al., 2003).

The correct approach is the isometric log-ratio (ILR) transformation with Helmert contrast matrices, which maps the interior of the simplex to an unconstrained $\mathbb{R}^{K-1}$ space where Gaussian distributions are appropriate (Pawlowsky-Glahn et al., 2015). We use Logistic-Normal ILR draws to propagate the race-bloc covariance Σ_Δ through the Monte Carlo, producing the 90% credible interval on $P(\text{win})$.

2.4 Demographic modeling and ecological inference

The ecological fallacy—inferring individual-level behavior from group-level aggregates—is central to demographic electoral modeling (Robinson, 1950; King, 1997). MRP addresses this by modeling individual vote probabilities as a function of individual-level covariates, then integrating over the population distribution (Park et al., 2004). We do not use MRP: it requires full joint cross-tabulations (race $\times$ religion $\times$ gender) that produce sparse cells at the intersection of minority groups, and its implementation is outside the scope of this project's timeline. Instead, we use three parallel marginal strata with an additive independence approximation and compare its outputs to an IPF-calibrated raked alternative.

3 Framework Architecture

3.1 Overview

The Electoral Equilibrium framework is a three-stage stochastic pipeline. Stage 1 translates a user-supplied shock narrative into per-stratum loyalty shift estimates. Stage 2 finds the coalition allocation that maximizes win probability under those shifts. Stage 3 continuously generates training data for Stage 1 from social media and news sources.

Each stage produces a frozen artifact (`ShockResponseData`, `EquilibriumData`, `SimulationData`) written to disk in a deterministic format keyed by a global random seed. All stochastic operations derive their seeds from a single top-level seed via `derive_seed`$(s_0, \text{stage\_name})$, ensuring byte-identical reproducibility across runs.

3.2 Demographic architecture: three parallel strata

The U.S. electorate is modeled as three independent marginal strata, each covering approximately 100% of voters:

- **Stratum 1 (Race/Ethnicity):** Five blocs—African American (≈12%), Latino (≈11%), Asian (≈5%), White (≈62%), and Other Race (≈10%)—constitute the optimizer's primary decision variables $\mathbf{w} \in \Delta^4$ (the 4-simplex).
- **Stratum 2 (Religion):** Seven blocs—Evangelical (≈24%), Catholic (≈21%), Protestant (≈13%), Secular (≈26%), Jewish (≈2%), Muslim (≈1%), and Other Religion (≈13%)—with fixed population weights $\mathbf{v}$ drawn from the voter panel.
- **Stratum 3 (Gender):** Three blocs—Women (≈52%), Men (≈47%), and Other Gender (≈1%)—with fixed population weights $\mathbf{g}$.

The three strata are explicitly not nested: we do not require race × religion × gender cross-tabulations. This avoids sparse-cell problems (e.g., Asian × Muslim × Other Gender ≈ 0.05% of the electorate) at the cost of an additive independence approximation discussed in Section 7.1.

3.3 Effective loyalty scalar

The three-stratum effective loyalty scalar is:

$$\mu_{\text{eff}}(\mathbf{w}) \;=\; \lambda_1 \sum_i w_i\, \mu_i^{(\text{race})} \;+\; \lambda_2 \sum_r v_r\, \mu_r^{(\text{rel})} \;+\; \lambda_3 \sum_g g_g\, \mu_g^{(\text{gen})}, \tag{1}$$

where $\mu_i^{(\text{race})}$, $\mu_r^{(\text{rel})}$, $\mu_g^{(\text{gen})}$ are the within-bloc Democratic vote shares estimated from the voter panel, and $\lambda_1 + \lambda_2 + \lambda_3 = 1$ with $\lambda_k \geq 0$. The optimizer controls only $\mathbf{w}$; the religion and gender weights $\mathbf{v}$ and $\mathbf{g}$ are fixed from the panel and are not decision variables.

The layer weights $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \lambda_3)$ are calibrated from historical data by iterative proportional fitting (Section 5.3). We maintain two calibrations in parallel: an *additive prior* (50/30/20) reflecting conventional wisdom about race's primacy in U.S. electoral politics, and a *raked* calibration (11.4/22.4/66.2) derived from historical two-party vote shares. The divergence between the two is itself a substantive finding (Section 6.2).

3.4 Win condition and Electoral College adjustment

The optimizer targets $\mu_{\text{eff}}(\mathbf{w}) \geq V_{\text{eq}}$, where V_{eq} is the popular-vote two-party share at which $P(\text{party wins EC}) = 0.50$. We derive V_{eq} by logistic regression:

$$P(\text{Dem EC win}) \;=\; \sigma(A \cdot x_{\text{dem2p}} + B)\,, \qquad V_{\text{eq}}^{(\text{Dem})} = -B/A, \tag{2}$$

where x_{dem2p} is the Democratic two-party popular vote share and σ is the logistic function. This is estimated on all 20 historical cycles, 1948–2024; see Section 6.1.

4 Data

4.1 Survey sources

The voter panel integrates six data sources across three strata:

1. **ANES Cumulative Data File (1948–2020):** American National Election Studies provides the core longitudinal panel for race, gender, and some religion variables. We use the VCF-coded cumulative file with auto-extracted label dictionary from the codebook PDF.

2. **Cooperative Election Study (CES, 2006–2024):** Over 700,000 respondents across six election cycles. CES is the primary source for religion blocs post-2006 and provides evangelical/Protestant disaggregation via the `race_h` Hispanic-override variable and evangelical flag column.

3. **General Social Survey (GSS, 1972–2022):** Primary source for religion bloc time series. Weights use the `wtssnrps` → `wtssps` → `wtss` fallback chain.

4. **National Election Pool Exit Polls (NEP, 2000–2024):** Gold standard for election-night demographic vote shares. Used as the primary source for race blocs in cycles where available and for benchmarking all panel estimates.

5. **NORC Public Opinion Research Survey (NPORS, 2024):** Race, religion, and gender vote shares for the 2024 cycle.

6. **Democracy Fund VOTER Panel:** Longitudinal voter tracking across multiple cycles, providing panel continuity.

4.2 Panel construction and conflict resolution

The panel covers 20 presidential election cycles (1948–2024) across 15 canonical demographic blocs (five race, seven religion, three gender), for a maximum of 300 bloc-cycle cells. After data collection, coverage was 234 cells (78.0%); all 15 blocs are present in every cycle from 2004 to 2024 (100% modern coverage).

When multiple sources report a vote share for the same bloc-cycle cell, conflicts are resolved by inverse-standard-error weighting:

$$\hat{v}_{\text{merged}} = \frac{\sum_s w_s \, v_s}{\sum_s w_s}, \qquad w_s = \frac{1}{\text{SE}_s}, \tag{3}$$

with SE estimated from the Kish effective sample size for ANES/GSS/CES and from the binomial formula for NEP. Each conflict is logged with per-source values.

4.3 Three-party contamination correction

Survey panels record the raw Democratic fraction of all votes cast, including third-party votes. In 1992, Ross Perot received 18.9% of the popular vote; Clinton's raw Democratic fraction (≈43%) substantially understates his two-party share (53.5%). We correct each 1992 bloc using bloc-specific Perot support from the 1992 National Exit Poll:

$$v_{i,1992}^{(2p)} = \frac{v_{i,1992}^{(\mathrm{raw})}}{1 - p_{i,1992}}, \tag{4}$$

where $p_{i,1992}$ is the Perot share for bloc i (White: 21%, African American: 7%, Latino: 14%, etc.). Without this correction, the 1992 panel row is misclassified as a Republican win and the covariance matrix Σ_Δ carries inflated race-bloc variance.

4.4 Structural gaps and imputation

Evangelical and secular religion blocs lack reliable data before 2000, when the relevant institutions (Moral Majority, "nones") either did not exist or were insufficiently represented in survey instruments. Muslim and Asian blocs are similarly sparse before the 1965 Immigration Act. We do not impute these structural gaps; instead, LLM fine-tuning for these blocs is restricted to 2004–2024 cycles where data exists. Three isolated cells (other_gender for five cycles, Muslim 2004, other_race 1948) are imputed from documented external sources with explicit source tags.

4.5 Panel quality benchmarks

Table 1 reports estimated Democratic vote shares from the panel alongside NEP and Pew benchmarks for the 13 Democratic winning cycles. All race blocs are within 7 percentage points of benchmarks; notable discrepancies are historically explained rather than indicative of data errors (see Section 7).

Table 1: Panel vote-share estimates vs. benchmark sources (Democratic winning cycles, 1948–2024).

Bloc	Panel μ	Benchmark	Deviation	Notes
African American	0.883	0.870	+1.3 pp	NEP
Latino	0.663	0.620	+4.3 pp	NEP
Asian	0.610	0.660	−5.0 pp	AAPI Data
White	0.456	0.390	+6.6 pp	NEP; ANES 2020 oversample
Other Race	0.644	0.580	+6.4 pp	NEP; heterogeneous bloc
Evangelical	0.275	0.230	+4.5 pp	Pew; pre-Moral-Majority cycles
Catholic	0.568	0.520	+4.8 pp	NEP
Secular	0.703	0.680	+2.3 pp	Pew
Jewish	0.789	0.720	+6.9 pp	Pew; CES education skew
Women	0.534	0.540	−0.6 pp	NEP
Men	0.487	0.430	+5.7 pp	NEP; pre-1980 gender gap

5 Methodology

5.1 Gaussian Process baseline classifier

The GP classifier validates that panel vote-share features are informative about realized election outcomes and provides a calibrated prior on $P(\text{win} \mid \text{coalition})$ for the optimizer.

5.1.1 Model specification

We fit a `GaussianProcessRegressor` on binary $\{0, 1\}$ outcomes using an additive kernel $k = k_{\text{RBF}} + k_{\text{Matérn}(\nu=1.5)}$. We use GPR rather than GPC because the paper requires cycle-level posterior standard deviations $\sigma_c = \text{std}[f(x_c)]$ as epistemic uncertainty estimates. The sklearn GPC (Laplace approximation) does not expose the latent posterior standard deviation through its public API; GPR returns it directly via `predict(return_std=True)`. GPR on binary labels is technically misspecified but manages this via:

- Predictions clipped to $[0, 1]$.
- Observation noise $\alpha = 0.10$, which models $\approx$3 pp survey measurement error in standardized label space and prevents exact interpolation through training points.

- Feature standardization via `StandardScaler` fitted on all available cycles before the leave-one-cycle-out (LOCO) loop.

Without `StandardScaler`, raw vote-share features (span $\approx$0.3–0.9) produce kernel length-scale collapse: the optimizer drives both RBF and Matérn length scales to their lower bounds (10^{-5}), reducing the model to a near-delta function and producing coin-flip accuracy.

5.1.2 Feature construction

Each training cycle is represented by a $d = 9$-dimensional feature vector:

- Five race-bloc Democratic vote shares $v_{i,c}^{(\text{race})}$.
- Three stratum-weighted coalition averages:

$$\begin{aligned} \bar{\mu}_c^{(\text{race})} &= \textstyle\sum_i \tilde{e}_i^{(\text{race})}\, v_{i,c}^{(\text{race})}, \\ \bar{\mu}_c^{(\text{rel})} &= \textstyle\sum_r \tilde{e}_r^{(\text{rel})}\, v_{r,c}^{(\text{rel})}, \\ \bar{\mu}_c^{(\text{gen})} &= \textstyle\sum_g \tilde{e}_g^{(\text{gen})}\, v_{g,c}^{(\text{gen})}, \end{aligned} \tag{5}$$

  where $\tilde{e}$ are approximate national electorate shares normalized over blocs present in cycle c.

- A normalized cycle-year feature $t_c = (c - 1948)/76 \in [0, 1]$, which gives the GP temporal ordering so that pre-realignment cycles (1948–1968) receive lower covariance with modern cycles.

The stratum-weighted averages were added after finding that the GP, presented with only five race-bloc features, fixated on African American loyalty in 2024 ($\mu_{\text{AA},2024} = 0.838$, matching a 2020 Democratic win pattern) while missing the coalition-wide deficit: $\bar{\mu}_{2024}^{(\text{race})} = 0.486$, $\bar{\mu}_{2024}^{(\text{rel})} = 0.439$, $\bar{\mu}_{2024}^{(\text{gen})} = 0.483$, all below $V_{\text{eq}} = 0.507$.

5.1.3 Leave-one-cycle-out validation

LOCO-CV trains on 19 of 20 cycles and predicts the held-out cycle. For each fold c, we refit the scaler and GP on the training set and report $\texttt{prob\_win}_c$ and $\texttt{prob\_std}_c$ (the GP posterior standard deviation) on the held-out cycle.

5.2 Moment estimation

The optimizer requires $\boldsymbol{\mu}^{(P)}$ (mean within-bloc Democratic vote share across winning cycles for party P) and Σ_Δ (the 5×5 empirical race-bloc covariance across all cycles).

Winning cycles are identified from the external ground-truth table `_PRES_DEM_2P_SHARE`, not from the panel-derived vote-share heuristic that fails in three-party years. The panel was verified to agree with certified two-party returns on 16 of 20 cycles; the four exceptions (1952, 1988, 1992, 2004) are corrected by the external table.

Covariance Σ_Δ is estimated from all 20 cycles (not conditioned on winning) using the sample covariance with ddof = 1. Ledoit-Wolf shrinkage (Ledoit & Wolf, 2004) is applied to guarantee positive-definiteness when the training set for a particular LOCO fold has fewer than $5 + 1$ observations:

$$\hat{\Sigma}_{\text{LW}} = (1 - \alpha)\, \hat{\Sigma}_{\text{sample}} + \alpha\, \hat{\mu}_{\text{sample}} I_{5\times 5}. \tag{6}$$

5.3 Layer weight calibration by iterative proportional fitting

We calibrate $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \lambda_3)$ by minimizing the MSE of μ_{eff} against historical two-party Democratic vote shares:

$$\hat{\boldsymbol{\lambda}} = \underset{\boldsymbol{\lambda}\in\Delta^2}{\arg\min} \sum_c \left(\lambda_1\, \bar{\mu}_c^{(\text{race})} + \lambda_2\, \bar{\mu}_c^{(\text{rel})} + \lambda_3\, \bar{\mu}_c^{(\text{gen})} - y_c\right)^2, \tag{7}$$

where y_c is the historical two-party Democratic popular vote share and $\Delta^2 = \{\boldsymbol{\lambda} : \sum_k \lambda_k = 1,\ \lambda_k \geq 0\}$.

The three stratum series $\bar{\mu}^{(\mathrm{race})}, \bar{\mu}^{(\mathrm{rel})}, \bar{\mu}^{(\mathrm{gen})}$ are nearly collinear—all track the partisan tide with pairwise correlations ≈ 0.92—making the unregularized objective nearly flat and causing first-order methods to converge extremely slowly: multiplicative IPF required ≈288 iterations; projected gradient descent required ≈101 iterations; FISTA (Nesterov momentum) required ≈35 iterations.

The correct approach is Tikhonov regularization toward an equal-weight prior:

$$\left(\mathbf{S}^{\top}\mathbf{S} + \rho I\right) \boldsymbol{\lambda} = \mathbf{S}^{\top}\mathbf{y} + \rho\, \boldsymbol{\lambda}_{\mathrm{prior}}, \tag{8}$$

where $\mathbf{S} \in \mathbb{R}^{n\times 3}$ stacks the stratum series, $\rho = 0.01$ is the regularization weight, and $\boldsymbol{\lambda}_{\mathrm{prior}} = (1/3, 1/3, 1/3)$ is the equal-weight prior. The 1% regularization reduces the condition number of $\mathbf{S}^{\top}\mathbf{S}$ from ≈120 to ≈12, allowing an exact one-shot solve followed by the Duchi et al. 2008 simplex projection. An active-set outer loop re-solves on the reduced simplex face if the projection zeros out a stratum; this typically requires one additional pass. The total iteration count is two.

5.4 Coalition optimizer (DQCP)

5.4.1 Objective formulation

The post-shock effective loyalty scalar is:

$$\tilde{\mu}_{\mathrm{eff}}(\mathbf{w}) \;=\; \lambda_1 \sum_i w_i \left(\mu_i + \delta_i^{(\mathrm{race})}\right) \;+\; \lambda_2 \sum_r v_r \left(\mu_r + \delta_r^{(\mathrm{rel})}\right) \;+\; \lambda_3 \sum_g g_g \left(\mu_g + \delta_g^{(\mathrm{gen})}\right), \tag{9}$$

where $\boldsymbol{\delta}$ are the shock deltas from Stage 1.

The optimizer maximizes the probability that $\tilde{\mu}_{\mathrm{eff}}(\mathbf{w})$ exceeds V_{eq} under a Gaussian model for the race-level shock uncertainty:

$$\max_{\mathbf{w}\in\Delta^4} \Phi\left(\frac{\tilde{\mu}_{\text{eff}}(\mathbf{w}) - V_{\text{eq}}}{\sqrt{\lambda_1^2\, \mathbf{w}^\top \Sigma_\Delta \mathbf{w}}}\right). \tag{10}$$

Since Φ is monotone increasing, this is equivalent to maximizing the argument. We prove below that the argument is quasi-convex in $\mathbf{w}$.

5.4.2 Quasi-convexity proof

Proposition 1. *The objective in (10) is quasi-convex in $\mathbf{w} \in \Delta^4$.*

Proof. Let $f(\mathbf{w}) = g(\mathbf{w})/h(\mathbf{w})$ where $g(\mathbf{w}) = \tilde{\mu}_{\text{eff}}(\mathbf{w}) - V_{\text{eq}}$ and $h(\mathbf{w}) = \sqrt{\lambda_1^2\, \mathbf{w}^\top \Sigma_\Delta \mathbf{w}}$.

Numerator. $g(\mathbf{w})$ is affine in $\mathbf{w}$ by inspection of (9): it equals $\lambda_1\, \boldsymbol{\mu}^{(\text{race})\top}\mathbf{w}$ plus terms that do not depend on $\mathbf{w}$. An affine function is simultaneously convex and concave.

Denominator. Σ_Δ is positive semi-definite by construction; Ledoit-Wolf shrinkage guarantees strict positive-definiteness. Therefore $\mathbf{w}^\top \Sigma_\Delta \mathbf{w}$ is a strictly convex quadratic form for $\mathbf{w} \in \text{int}(\Delta^4)$, and $h(\mathbf{w}) = \lambda_1\sqrt{\mathbf{w}^\top \Sigma_\Delta \mathbf{w}}$ is convex (composition of a convex function with $\sqrt{\cdot}$, which is concave and monotone increasing on $\mathbb{R}_+$, preserves convexity). Moreover $h(\mathbf{w}) > 0$ for all $\mathbf{w}$ with $w_i > 0$ for some i.

Ratio. By Proposition 1 of Diamond & Boyd (2019), a ratio of an affine function to a positive convex function is quasi-convex. Therefore $f(\mathbf{w})$ is quasi-convex on Δ^4.

Composition. Φ is monotone increasing, and the composition of a quasi-convex function with a monotone increasing function is quasi-convex. Therefore $\Phi(f(\mathbf{w}))$ is quasi-convex, and its maximization is a quasi-convex program (QCP). □ □

The program is solved by CVXPY with `problem.solve(qcp=True)`; the implementation asserts `problem.is_dcp(qcp=True)` at runtime as a correctness guard.

Why not min-variance? The standard mean-variance portfolio objective $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma_\Delta \mathbf{w}$ subject to $\boldsymbol{\mu}^\top \mathbf{w} \geq V_{\text{eq}}$ is equivalent to asking: "minimize uncertainty given the win constraint." When $\boldsymbol{\mu}^\top \mathbf{w}$ already exceeds V_{eq} for the high-loyalty blocs, this approach concentrates weight there and assigns zero weight to blocs that add variance without a proportional loyalty gain. In our Democrat baseline, this produces $w_{\text{Latino}} = w_{\text{Asian}} = 0$ even though both groups represent substantial electorates. The quasi-convex objective—maximizing $P(\text{win})$ rather than minimizing variance—restores positive weight to these blocs in deficit scenarios where the path to V_{eq} requires accepting additional variance.

5.5 Monte Carlo uncertainty quantification

The 90% credible interval on $P(\text{win})$ is computed via 10,000 Logistic-Normal ILR draws. Let $\mathbf{w}^*$ be the optimizer solution.

1. Map $\mathbf{w}^*$ to ILR coordinates: $\mathbf{z}^* = \mathbf{V}^\top \log(\mathbf{w}^*)$, where $\mathbf{V} \in \mathbb{R}^{5\times 4}$ is the Helmert contrast matrix.
2. Propagate Σ_Δ to ILR space via the Jacobian: $\Sigma_{\text{ILR}} = J\,\Sigma_\Delta\,J^\top$.
3. Draw $\mathbf{y}^{(n)} \sim \mathcal{N}(\mathbf{z}^*, \Sigma_{\text{ILR}})$ for $n = 1, \ldots, 10{,}000$.
4. Back-transform: $\mathbf{w}^{(n)} = \text{softmax}(\mathbf{V}\mathbf{y}^{(n)})$.
5. Compute the win flag $\mathbf{1}[\mu_{\text{eff}}(\mathbf{w}^{(n)}) \geq V_{\text{eq}}]$ and the 5th and 95th percentiles as the 90% CI bounds.

The Dirichlet distribution was explicitly rejected for this step because its covariance structure forces $\text{Cov}(w_i, w_j) < 0$ for all $i \neq j$, making it impossible to model wave elections where all blocs shift together. The delta method with a floor at $w_i \geq 0.01$ was rejected because the perturbation is not uniform in Aitchison geometry.

5.6 Stage 1: LLM fine-tuning for shock estimation

Stage 1 translates a user-supplied political shock narrative into per-stratum loyalty shift bins. We fine-tune Mistral 7B Instruct v0.3 (Jiang et al., 2023) via QLoRA (rank 16, $\alpha = 32$, 4-bit NF4 quantization) on a catalog of historical shock events.

Output vocabulary. The model is constrained to produce exactly one of nine tokens per bloc: {`strong_neg`, `mod_neg`, `mild_neg`, `slight_neg`, `neutral`, `slight_pos`, `mild_pos`, `mod_pos`, `strong_pos`}, implemented via the `outlines` constrained decoding library (Willard & Louf, 2023). Token midpoints span $[-0.12, +0.12]$ in vote-share space. Per-bin uncertainty σ_b is empirically estimated from historical variation and stored in `configs/bin_uncertainty.json`; conservative priors are used for bins with fewer than five historical examples.

Training data. Historical training records pair shock descriptions with per-stratum delta bins derived from pre/post-event changes in survey vote shares. Synthetic data is generated via Gemini 2.5 Pro and validated by three diagnostics before inclusion: (i) maximum mean discrepancy (MMD) with RBF kernel and the median heuristic $\lambda = 1/(2\sigma^2)$ (Gretton et al., 2012); (ii) PCA alignment requiring $> 70\%$ of synthetic variance explained by the top-2 real principal components; (iii) pairwise correlation difference (PCD) Frobenius norm.

5.7 Stage 3: NLP pipeline

Stage 3 generates training data for Stage 1 from social media and news archives. Author demographic inference uses a three-stage pipeline: (i) a SetFit few-shot classifier (Tunstall et al., 2022) on bio text, served from a Raspberry Pi 5 running `all-MiniLM-L6-v2` via sentence-transformers in CPU mode (≈20 ms per bio; Hailo NPU compilation deferred pending vendor SDK availability); (ii) a keyword lexicon fallback (`configs/race_lexicon.json`, `religion_lexicon.json`, `gender_lexicon.json`); and (iii) a language-prior fallback based on Pew Research population estimates. Posts assigned via the language prior are excluded from both the mean $\boldsymbol{\mu}$ and covariance Σ_Δ estimation—including them in only one would bifurcate the data-generating process and violate the mean-variance optimizer's assumptions.

Sentiment toward each demographic bloc is scored by a bloc-weighted RoBERTa sentence encoder (Liu et al., 2019). The resulting ElasticityScore scalars $\in [-1, 1]$ serve as input features to the Stage 1 LLM alongside the free-text shock description.

6 Empirical Results

6.1 Electoral College threshold derivation

Fitting the logistic regression in (2) on all 20 historical cycles yields:

$$V_{\text{eq}}^{(\text{Dem})} = 0.5066, \qquad V_{\text{eq}}^{(\text{Rep})} = 0.4934. \tag{11}$$

Democrats require 50.66% of the two-party popular vote for a 50% probability of winning the Electoral College; Republicans require only 49.34%—a 1.32 percentage-point geographic efficiency advantage from their concentrated distribution in rural and small-swing states. The 2000 (Gore: 50.3%, Bush EC win) and 2016 (Clinton: 51.1%, Trump EC win) mismatches are the primary empirical anchors driving the Democrat threshold above 50%.

Substituting the flat 50% threshold used in some prior work would systematically understate the Democrat structural disadvantage. We recommend deriving V_{eq} from EC-adjusted logistic regression for any coalition model targeting the Electoral College.

6.2 Layer weight calibration

Table 2 reports the additive prior and IPF-calibrated (raked) layer weights.

Table 2: Layer weight calibration: additive prior versus raked (IPF) values. Raked weights are derived by minimizing MSE of μ_{eff} against historical two-party Democratic vote shares with 1% Tikhonov regularization.

	Race (λ_1)	Religion (λ_2)	Gender (λ_3)
Additive prior	0.500	0.300	0.200
Raked (IPF)	**0.114**	**0.224**	**0.662**

The large gender weight ($\lambda_3 = 0.662$) is the primary finding. The gender gap in Democratic presidential vote share was near zero in the 1950s—Eisenhower won women by 3 pp in 1956—and grew to approximately +12 pp by 2020. This cross-cycle growth gives gender the highest temporal variance of the three strata and correspondingly the highest explanatory power in the least-squares regression. Race's low raked weight ($\lambda_1 = 0.114$) reflects that African American Democratic loyalty has been 87–92% since 1964: consistently high, but low temporal variance contributes little to the regression.

Remark 1. *The additive prior and the raked values answer different questions. The additive prior (50/30/20) reflects electoral campaign intuition: which stratum matters most for resource allocation? Race commands the largest share because the mobilizable margins among African American and Latino voters are largest. The raked calibration (11/22/67) answers a different question: which stratum has explained the most historical variation in Democratic vote share? The optimizer must decide which objective is appropriate for the prescriptive task of coalition rebalancing; we present both and compare their outputs.*

6.3 GP classifier validation

Table 3 shows the sequential improvement of the GP classifier as data quality and feature engineering issues are resolved.

Table 3: GP classifier LOCO-CV results across successive model versions (Democrat, 20 cycles 1948–2024). Brier score is the primary calibration metric; accuracy is reported for comparability with the forecasting literature.

Version	Fix applied	Acc.	Brier	$p(\text{Dem}, 2024)$	Misses
V0	Baseline (raw features)	0.500	0.254	—	(coin flip)
V1	+StandardScaler	0.800	0.122	0.806	1992, 2000, 2004, 2024
V2	+Perot correction (1992)	0.850	0.113	0.647	2000, 2024
V3	$+\alpha = 0.10$	0.900	0.090	0.647	2000, 2024
V4	+Stratum features	0.850	**0.071**	0.542	1988, 2000, 2024

The final model (V4) achieves Brier score 0.071 and 85% LOCO accuracy. The Brier improvement from V3 to V4 is 21%; the accuracy drop of one cycle (1988) is attributable to a known ANES oversampling artifact rather than model error (Section 7.2).

Remaining misses.

- *2000* ($p = 0.479$, $y = 1$): Gore won the two-party vote by 0.5 pp. The model correctly signals near-maximum epistemic uncertainty ($\sigma_{2000} = 0.38$). This cycle is irreducibly borderline.

- *2024* ($p = 0.542$, $y = 0$): Harris lost the two-party popular vote by 0.8 pp. The model predicts a slight Democratic lean ($p = 0.542$) with $\sigma_{2024} = 0.238$. The nearest temporal neighbor is 2020 ($y = 1$, $\mu_{\text{eff}} = 0.532$); with a smooth Matérn kernel and only 20 training cycles, the model cannot learn a sharp decision threshold at $V_{\text{eq}} = 0.507$. The Brier contribution from 2024 improved from $(0.647)^2 = 0.419$ (V3) to $(0.542)^2 = 0.294$ (V4). We interpret $p = 0.542$ with $\sigma = 0.238$ as a correctly calibrated toss-up prediction for a 0.8 pp loss.

Feature standardization as a reproducibility warning. The single most impactful fix—from 50% to 80% accuracy—was `StandardScaler` (Table 3, V0 $\rightarrow$ V1). Without standardization, raw vote-share features (range ≈0.3–0.9 across blocs) cause kernel optimizer collapse. The resulting model trains silently and produces random-baseline predictions with no visible error message. This failure mode will recur in any subsequent application of GP classifiers to vote-share data; we document it explicitly.

Platt scaling. We implemented jackknife Platt scaling as a post-hoc calibration step and found that it degraded every metric (accuracy: $0.800 \rightarrow 0.750$; Brier: $0.122 \rightarrow 0.162$; 2024 market divergence: 12.9 pp $\rightarrow$ 31.2 pp). The diagnosis is that the apparent "calibration error" is not a calibration error: GP features are drawn from post-election voter panel data, while prediction markets set prices on pre-election polling. The GP correctly assigns high confidence to cycles with unambiguous panel vote shares; that confidence is appropriate given the information state. Comparing GP outputs to market prices is informative as a consistency check, not as a calibration audit.

6.4 Baseline portfolio

The min-variance CVXPY baseline is:

$$\min_{\mathbf{w}\in\Delta^4} \mathbf{w}^\top \Sigma_\Delta \mathbf{w}, \quad \text{subject to } \boldsymbol{\mu}^{(\text{race})\top}\mathbf{w} \geq V_{\text{eq}}. \tag{12}$$

Table 4: Democrat baseline coalition from min-variance QP ($V_{\text{eq}} = 0.5066$). μ is the winning-cycle mean vote share; σ is the historical standard deviation.

Bloc	Weight w_i	μ_i	σ_i	μ_i/σ_i
African American	0.510	0.892	0.104	8.58
White	0.302	0.467	0.073	6.40
Other Race	0.188	0.579	0.233	2.49
Latino	0.000	0.669	0.106	6.31
Asian	0.000	0.579	0.245	2.36
$\mu_{\text{eff}} = 0.659$		Margin above V_{eq}: +12.4 pp		

The min-variance optimizer assigns zero weight to Latino and Asian voters (Table 4). Both blocs are electorally substantial (24M and 10M registered voters respectively) and moderately Democratic, but their per-cycle variance exceeds what the optimizer can justify given the loyalty already achieved by concentrating on African Americans (highest loyalty-to-variance ratio) and White voters (lowest absolute variance). This result is mathematically correct for the min-variance criterion. It is the paper's primary motivation for the DQCP objective: only by maximizing $P(\text{win})$ rather than minimizing variance does the optimizer confront deficit scenarios in which the path to V_{eq} necessarily runs through high-variance blocs.

7 Discussion

7.1 The ecological fallacy and additive independence

The additive independence assumption in (1) is an acknowledged approximation of the ecological fallacy (Robinson, 1950). A Latino Evangelical man's political behavior is not reconstructed by summing three marginal averages. The correct individual-level model is MRP (Park et al., 2004), which requires full race × religion × gender cross-tabulations. At intersection cells involving minority groups (Asian × Muslim, $n \approx 0.05\%$ of the electorate), survey sample sizes in available data produce unreliable estimates.

We take two steps to mitigate the approximation. First, raking (IPF across all three marginal tables) enforces joint consistency without requiring full joint distributions. The divergence between additive and raked outputs quantifies the degree of intersectional interaction the additive model misses. Second, the paper compares optimizer outputs under both $\boldsymbol{\lambda}$ calibrations; substantial divergence (> 2 pp in μ_{eff}) would indicate that intersectional interactions are load-bearing and MRP should be pursued in a follow-on study.

7.2 Data quality issues and their corrections

1988 ANES oversampling. Adding religion and gender stratum features (V4) caused the 1988 cycle to flip from correctly classified (model: $p = 0.45$, truth: Republican win) to misclassified ($p = 0.510$). The ANES 1988 panel shows $\mu_{\text{eff}} = 0.521$—which looks like a Democratic coalition—while Dukakis lost 46.1% of the two-party popular vote. The ANES 1988 survey is known to oversample civically engaged Democrats; the religion and gender strata amplify the overestimate relative to the race-only model. We do not apply a correction because no principled method exists to adjust a survey overestimate without access to the original sampling frame. The 1988 mis-classification is documented as a panel data artifact; the Brier score improvement across all other cycles ($0.090 \to 0.071$) justifies retaining the stratum features.

White voter baseline. Panel $\mu_{\text{white}} = 0.456$ is 6.6 pp above the NEP gold standard of 39%. The excess reflects two compounding effects: (i) winning cycles include 1948–1964, when white Southern Democrats were reliably Democratic before the post-1968 realignment; and (ii) ANES 2020 lacks post-stratification weights (the weight column is fully null), producing equal-weighted estimates that over-represent educated white Democrats. Papers reporting white-bloc loyalty estimates should restrict to post-1980 cycles or note the historical drift explicitly.

Asian covariance inflation. $\sigma_{\text{Asian}} = 0.245$, roughly 3.4× the white variance ($\sigma_{\text{white}} =$

0.073). Pre-1965 ANES Asian sub-samples are extremely small; the 1965 Immigration and Nationality Act fundamentally changed the Asian American electorate's size and composition; and the 1992 Perot correction uses a national-average proxy for Asian Perot support (15%) rather than a bloc-specific estimate. The inflated variance will propagate into Monte Carlo confidence intervals when the optimizer allocates weight to the Asian bloc; results for that scenario should be treated with corresponding caution.

7.3 Prediction market calibration

Post-shock model win-probability estimates are compared against Polymarket and PredictIt contract prices. We use prediction markets as post-hoc calibration benchmarks only—never as training inputs. Two reasons:

1. $\Delta\pi \neq \Delta\mu$: a change in the market's win-probability estimate aggregates traders' beliefs about the full electorate, whereas $\Delta\mu_{\text{eff}}$ is a structural demographic shift. Treating market price changes as supervision for a demographic shift model conflates two distinct quantities.

2. Feature timing: the GP classifier operates on post-election voter panel data. When predicting 2020 in LOCO-CV, the GP uses the 2020 panel vote shares—which record what voters actually did. A market price on election day 2020 (PredictIt: 65% Biden) reflects pre-election polling uncertainty, not the realized demographic outcome. Calibrating against market prices in this setting suppresses genuine signal.

The market comparison is presented in the paper as a consistency check: do the model and markets agree on direction? Large magnitude divergences in post-hoc review flag potential feature-timing or data quality issues for investigation. In the 2024 case, the 7.2 pp divergence between the model ($p = 0.542$) and PredictIt (47.0% Democrat) is within the noise level of a borderline election and does not indicate model failure.

7.4 Era sensitivity and the pre-realignment panel

We tested whether restricting training data to post-realignment cycles (1980+, 1992+) improves accuracy by removing the structural discontinuity of white Southern Democrats before 1968. The results were unambiguous:

Training era	n	Accuracy	Brier	$p(\text{Dem}, 2024)$
1948–2024 (full)	20	**0.900**	**0.090**	0.647
1980–2024	12	0.750	0.144	0.619
1992–2024	9	0.667	0.235	0.871

Restricting to modern cycles makes every metric worse. The GP's year feature gives temporal ordering to the kernel, so pre-realignment cycles contribute low-covariance training signal rather than noise. The full seven-decade panel is a genuine asset, not a liability, because the year feature contextualizes the demographic drift rather than allowing it to contaminate modern predictions.

8 Conclusion

We have presented Electoral Equilibrium, a stochastic coalition optimization framework that models the prescriptive problem of post-shock coalition rebalancing. The framework's key methodological contributions are: (i) an Electoral College–adjusted win threshold derived from logistic regression on historical cycles; (ii) an IPF-calibrated three-stratum layer weight model that recovers the known gender-gap trend from first principles; (iii) a quasi-convex Sharpe-ratio coalition optimizer that avoids the systematic exclusion of minority blocs that afflicts min-variance approaches; and (iv) a Logistic-Normal ILR Monte Carlo that correctly models wave elections and does not require a compositional floor.

The GP baseline classifier achieved 85% LOCO accuracy and Brier score 0.071 on the 20-cycle panel, demonstrating that voter panel survey data alone—without polling, prediction markets, or news signals—carries substantial retrospective signal. The four sequential fixes required to reach this accuracy (feature standardization, Perot three-party correction, observation noise regularization, and stratum-weighted features) are documented as reproducibility guidance for subsequent researchers applying GP models to vote-share data.

The large divergence between the additive prior ($\boldsymbol{\lambda} = (0.50, 0.30, 0.20)$) and the raked calibration ($\boldsymbol{\lambda} = (0.11, 0.22, 0.66)$) is not an artifact but a substantive finding: race dominates coalition *strategy* because mobilizable margins are largest there, while gender dominates *historical variation* because the gender gap has grown the most since 1980. Future work will use the DQCP optimizer to demonstrate that these two objectives recommend different coalition allocations under identical shock conditions.

The full framework—LLM shock estimation, DQCP optimizer, and ILR Monte Carlo—is not yet complete. Future stages will: (i) assemble a catalog of 25+ historical shock events; (ii) fine-tune Mistral 7B via QLoRA on the CMC HPC A100; (iii) collect social media and news data via the described NLP pipeline; and (iv) run the complete end-to-end pipeline on 2024 post-election shock scenarios to validate that the optimizer's recommended rebalancing is consistent with observed party strategic responses.

A Proof of Simplex Projection Complexity

The Duchi et al. 2008 simplex projection algorithm maps $\mathbf{v} \in \mathbb{R}^n$ to $\Pi_{\Delta^{n-1}}(\mathbf{v})$ in $\mathcal{O}(n \log n)$ time. For our calibration problem ($n = 3$), the algorithm is equivalent to a three-element sort followed by a closed-form threshold computation. We use the general $\mathcal{O}(n \log n)$ implementation for correctness in the domain-agnostic case.

The algorithm: sort $\mathbf{v}$ descending as $u_1 \geq \cdots \geq u_n$; compute cumulative sums $\rho = \max\{j : u_j > (\sum_{k=1}^{j} u_k - 1)/j\}$; set $\theta = (\sum_{k=1}^{\rho} u_k - 1)/\rho$; return $\max(v_i - \theta, 0)$ for each i.

B Convergence of Raking Algorithm

The three failed approaches before the regularized normal equations:

1. **Multiplicative IPF (coordinate descent):** ≈288 iterations to tolerance 10^{-6}. O(1/t) convergence rate on the near-flat landscape.

2. **Projected gradient descent:** ≈101 iterations. Same O(1/t) rate; full-gradient steps do not accelerate convergence on a nearly-flat objective.

3. **FISTA (Nesterov momentum):** ≈35 iterations. $O(1/t^2)$ rate improves over gradient descent by ≈3×, but the large condition number (≈120) of $\mathbf{S}^\top\mathbf{S}$ limits practical convergence speed.

The regularized normal equations reduce the condition number to ≈12 and allow an exact one-shot solve. Sensitivity analysis shows that varying ρ from 0.005 to 0.02 shifts the raked $\boldsymbol{\lambda}$ by less than 0.01 in any component.

C LOCO-CV Fold Results

Table 5: LOCO-CV fold results: Democrat, V4 model (stratum features, $\alpha = 0.10$, StandardScaler, Perot correction).

Cycle	p(win)	σ	y_{true}	Correct?	Notes
1948	0.613	0.471	1	✓	Truman
1952	0.341	0.496	0	✓	Eisenhower
1956	0.298	0.503	0	✓	Eisenhower
1960	0.671	0.458	1	✓	Kennedy
1964	0.812	0.372	1	✓	Johnson
1968	0.339	0.487	0	✓	Nixon

Cycle	p(win)	σ	y_{true}	Correct?	Notes
1972	0.204	0.511	0	✓	Nixon
1976	0.558	0.481	1	✓	Carter
1980	0.287	0.493	0	✓	Reagan
1984	0.261	0.497	0	✓	Reagan
1988	0.510	0.484	0	×	ANES oversample
1992	0.837	0.391	1	✓	Clinton (Perot-corrected)
1996	0.627	0.462	1	✓	Clinton
2000	0.479	0.382	1	×	Gore (+0.5 pp, borderline)
2004	0.453	0.401	0	✓	Bush
2008	0.782	0.321	1	✓	Obama
2012	0.734	0.344	1	✓	Obama
2016	0.591	0.457	1	✓	Clinton (popular win)
2020	0.703	0.388	1	✓	Biden
2024	0.542	0.238	0	×	Harris (−0.8 pp, borderline)
Accuracy: 17/20 = 0.850 Brier: 0.071					

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